

Sai Akshay Sivakumar

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EDUCATION

University of Maryland, College Park

College Park, MD

Bachelor of Science in Computer Science, Machine Learning Concentration; Minor in Statistics

Expected May 2026

- Relevant Coursework: Data Structures, Algorithms, Computer Systems, Databases, Artificial Intelligence, Machine Learning, Computer Vision, Linear Algebra, Statistics

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, OCaml, HTML/CSS, MATLAB, R, SAS

Tools & Frameworks: Linux, Docker, Kubernetes, Git, CMake, Spack, AWS, Jupyter, Pandas, NumPy, scikit-learn, React, Next.js, JUnit, Matplotlib, Bootstrap

Security Clearance: Active Secret Clearance

EXPERIENCE

Lockheed Martin

June 2025 – August 2025

Software Engineer Intern

Orlando, FL

- Built an event-driven C++ service to ingest and process real-time device events, implementing custom handlers for 5+ operations and enabling low-latency display of operational data in a user-facing interface.
- Extended a large-scale CMake build system with custom targets that centralized build metadata across 10+ applications, reducing configuration errors by 25% and improving automation and traceability in the build pipeline.
- Containerized Linux build workflows with Docker and Spack to standardize dependency management for 20+ C++ libraries and packages, improving environment reproducibility and reducing developer setup time by 30%.
- Delivered 13+ merge requests in an Agile environment, contributing code reviews, documentation, and maintainable changes aligned with sprint goals.

Headstarter AI

July 2024 – August 2024

Software Engineer bootcamp student

Remote

- Built a full-stack item-tracking application using Next.js, React, and OpenAI APIs, designing REST endpoints for item creation, search, and recommendation workflows with 95% recommendation accuracy in testing.
- Developed an AI customer support prototype integrating Next.js, React, WebSpeech API, and a Python sentiment analysis service; tested 20+ query scenarios, achieved 92% test-user satisfaction, and added automated tests for response quality and error handling.

University of Maryland

August 2022 – December 2023

Undergraduate Researcher

College Park, MD

- Built a campus-scale routing application using JavaScript, Python, HTML/CSS, and OpenStreetMap data to compute optimal walking paths across 257 buildings on a 1,340-acre campus.
- Implemented and optimized a Dijkstra-based pathfinding algorithm that improved average route computation time by 238% while handling 10,000+ graph nodes and edges with walkway, building, and access constraints.
- Automated multi-variable optimization experiments across 100+ parameter combinations using MATLAB and Excel workflows to evaluate tradeoffs between competing objectives.

PROJECTS

Race Simulator | *Python, Bayesian Networks, Particle Filters, Kalman Filters, Git*

- Built a Python race simulation and state-estimation engine using a particle filter to estimate vehicle position and orientation on a virtual track with over 90% localization accuracy under noisy sensor conditions.
- Implemented a Bayesian network to model race dynamics and driver behavior, enabling probabilistic prediction of race outcomes and pit-stop strategies.
- Developed a Kalman filter module to estimate vehicle velocity and location from noisy GPS-like inputs, reducing average prediction error by 35% compared with a naive smoothing baseline.

UFC Judge | *Python, Pandas, SQL, scikit-learn, Jupyter, UFC API*

- Built a Python-based ML system to analyze round-level fight statistics and improve scoring consistency by 20% relative to a baseline human-scoring benchmark.
- Integrated the official UFC API to ingest and normalize historical and live fight data, building reusable data pipelines for downstream analysis and model training.
- Used Pandas and SQL to analyze relationships between striking, grappling, and control metrics and round-level judging outcomes, then trained and evaluated a logistic regression model to generate more stable, data-driven scorecards.